

Are data centres harming bees? Scientists are investigating the impact of electromagnetic fields | - The Times

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Bees are among the world's most important pollinators, helping sustain ecosystems and supporting the production of many fruits, vegetables, nuts and seeds. Yet bee populations have been declining for years due to habitat loss, pesticides, diseases, parasites and climate change. More recently, another potential threat has entered the conversation: the rapid expansion of data centres. Some beekeepers living near large facilities have reported fewer bees and weaker colonies, raising concerns that electromagnetic fields (EMFs) generated by high-powered electrical infrastructure could interfere with the insects' natural behaviour. While researchers agree that the issue deserves further study, they also caution that the scientific evidence is still evolving and does not yet establish data centres as a confirmed cause of widespread bee declines. Can electromagnetic fields from data centres affect bees? Honeybees rely on multiple cues to navigate, including the position of the Sun, landmarks, odours and the Earth's magnetic field. Numerous laboratory studies have explored whether artificial electromagnetic fields generated by power lines, mobile communication networks and other electrical infrastructure can interfere with these natural navigation systems. A scientific review by Environmental Health Sciences examined research on the effects of anthropogenic electromagnetic fields on insects. The review found evidence that exposure to certain EMF frequencies and intensities has been associated with changes in insect behaviour, reduced reproductive success, oxidative stress, DNA damage and altered development under experimental conditions. However, the authors also noted that many studies differed significantly in methodology, making it difficult to draw universal conclusions about real-world impacts. Scientists emphasise that the effects observed in controlled laboratory settings do not necessarily translate directly to environments around operational data centres, where exposure levels can vary considerably. Why are some beekeepers linking data centres to bee declines? Concerns have grown after some farmers and beekeepers reported noticeable changes in bee activity following the construction of nearby data centres. Some have described fewer bees visiting flowers, smaller colonies and changes in foraging behaviour after large electrical facilities began operating nearby. While these observations remain anecdotal and cannot, by themselves, establish a causal relationship, they have drawn attention to a growing body of research examining how anthropogenic electromagnetic fields (EMFs) may affect insects. A comprehensive 2023 review published in *Science Advances* analysed many studies investigating the effects of human-generated electromagnetic radiation on insects, including bees, butterflies and flies. The researchers found that almost 60% of the studies reported at least one statistically significant biological effect, ranging from altered behaviour and reduced reproductive success to oxidative stress, DNA damage

and changes in development. The review concluded that there is “substantial evidence that anthropogenic EMFs can affect insects”, although it also noted that the quality and methodologies of existing studies vary considerably and that more standardised research is needed before broad conclusions can be drawn. For bees specifically, previous laboratory and field studies have suggested that electromagnetic fields may interfere with magnetoreception, the ability to detect the Earth’s magnetic field, which honeybees use alongside the Sun, landmarks and odours to navigate. Researchers have also observed changes in waggle dance communication, reduced return rates of foraging bees, altered colony behaviour and physiological stress under certain experimental conditions. However, the intensity, duration and frequency of EMF exposure differ widely across studies, making it difficult to determine whether similar effects occur around real-world infrastructure such as data centres. The authors of the Science Advances review emphasised that the rapid expansion of wireless technologies and electrical infrastructure has outpaced research into their ecological consequences. They wrote that “there is evidence of diverse influences of anthropogenic electromagnetic radiation on insects, with consequences that may scale up to affect ecosystem services,” while also calling for more rigorous long-term field studies to understand how these exposures interact with other stressors affecting insect populations. Scientists caution that bee declines remain a multifactorial issue. Parasites such as the Varroa destructor mite, viral diseases, pesticide exposure, habitat loss, climate change and poor nutrition are all well-established drivers of colony losses. As data centres continue to expand worldwide to support artificial intelligence and cloud computing, researchers say carefully designed field investigations will be essential to determine whether electromagnetic emissions from such facilities represent an additional environmental pressure on pollinators, or whether other factors better explain the observations reported by beekeepers. Why protecting bees remains critical regardless of the cause Whether or not data centres ultimately prove to be a significant factor, there is broad scientific agreement that pollinators are under increasing pressure. According to the Apiary Insects of America and bee industry surveys, U.S. beekeepers reported losing an estimated 55.6% of managed honeybee colonies during 2024-25, the highest annual loss recorded in the survey’s history. 75% of the world’s leading food crops benefit to some extent from animal pollination, while pollinators contribute significantly to global agricultural production and biodiversity. Millions of livelihoods also depend on healthy pollinator populations, particularly among smallholder farmers. Researchers say protecting bees requires addressing multiple threats simultaneously by restoring wildflower habitats, reducing harmful pesticide exposure, improving disease management, supporting sustainable beekeeping and continuing research into emerging environmental pressures, including the possible effects of electromagnetic fields. As new technologies reshape landscapes, understanding how they interact with wildlife will be essential for balancing technological development with environmental conservation.